Per-Operator Body and Hand Calibration for MABEL VR Teleoperation:

Measuring the Operator from Sparse Head + Wrist + Finger Tracking, and Using It to Scale Retargeting, Anchor the Shoulder, and Size the Hands

MABEL Project — Control & Teleoperation

November 2, 2026

Abstract

MABEL's VR teleoperation maps an operator's head pose, two wrist poses, and per-finger hand skeleton (from an Apple-Vision-Pro-class headset) onto a mobile bimanual humanoid. The legacy mapping used a single fixed workspace gain and re-estimated the operator's finger lengths live every solve, so a tall and a short operator covered the robot's workspace differently and the fingers tracked noisily. This document specifies a one-time, session-start *calibration* that measures the operator's anthropometry from the same sparse signals the headset already streams—stature, arm length, shoulder width, shoulder-joint centres, neck height, per-finger bone lengths, hand size, and pinch span—and feeds them into the retargeting scale, an optional shoulder-anchored mapping, the per-finger hand retargeter, and the grasp curve. We give the operator body-frame construction, the three calibration poses, and the closed-form anthropometry, all grounded in the canonical Winter/Drillis–Contini segment ratios and the sparse-input VR-teleop literature. Experiments on synthetic operators (1.45 m–2.07 m) show exact dimension recovery in the noise-free case and ≤ 1.8 mm worst-case error at 5 mm tracker noise, and show that the fixed gain over/under-drives the workspace by 35 % across operator sizes while the per-user gain holds it flat. The calibration is a *refinement*: it preserves the hand-tuned feel for a reference operator and only corrects for size.

Keywords: teleoperation, motion retargeting, anthropometric calibration, cross-embodiment, workspace scaling, dexterous hand retargeting, Apple Vision Pro.

1 Overview

A teleoperator wearing a head-mounted tracker provides only three rigid poses (head and two wrists) plus a hand skeleton—no torso, shoulders, or elbows. Different operators have different proportions, so the same wrist motion should map to different robot targets. We resolve this with a short calibration that condenses the operator into an `OperatorProfile` (`mabel/whole_body_retargeting/calibration.py`) consumed by the retargeter (`mabel/whole_body_retargeting/wrist_clutch.py`), the per-finger hand retargeter (`mabel/hand_retargeting/keypoint_ik.py`, `mabel/hand_retargeting/vector.py`), and the grasp curve. The design follows the two camps of the sparse-input literature: per-user *body-size fitting* (EgoPoser [1], QuestSim [2]) and *workspace scaling* for cross-embodiment teleop (ACE [3], Open-TeleVision [4], AnyTeleop [5]).

2 Method

2.1 Operator body frame

All geometry is expressed in a yaw-only, gravity-aligned *operator body frame* so that head pitch/roll never contaminate a "lateral" or "vertical" measurement. Given the live head pose $H = \left[\begin{smallmatrix} R_h & p_h \\ 0 & 1 \end{smallmatrix}\right]$ in the ARKit world frame ($+y$ up, $-z$ forward), let $u = (0, 1, 0)^\top$ be world up. The head heading $-R_h e_3$ is flattened to the horizontal plane and orthonormalised:

$$\hat{f} = \frac{(-R_h e_3) - \big((-R_h e_3) \cdot u\big)\, u}{\|\cdot\|}, \qquad \hat{r} = \hat{f} \times u, \qquad B = [\,\hat{r} \;\; u \;\; \hat{f}\,] \in \mathbb{R}^{3\times 3}. \tag{1}$$

The body coordinates of a world point p are $b = B^\top (p - p_h) = (\text{right}, \text{up}, \text{forward})$. B is the single source of truth shared by the calibrator and the shoulder anchor (`calibration.body_frame`).

2.2 The three calibration poses

Each pose is auto-captured once the tracked points (head + both wrists) hold still—defined by low positional *spread* over a 0.25 s window, not frame-to-frame speed, which is dominated by tracker jitter—for 0.5 s, then averaged over a 0.8 s window. A motion gate requires the operator to move between poses so leftover still frames of one pose are never mis-attributed to the next.

1. T-pose (arms horizontal). With the arms straight out to the sides, the shoulder shares each wrist's height and forward offset (the arm is purely lateral), so the per-side wrist body coordinates b_ℓ^T, b_r^T localise the shoulder in two of three axes, and the lateral coordinate carries the arm length. The world wrist separation gives the arm span.

2. Arms-forward (arms straight ahead). With both arms pointing forward and parallel, the lateral (body-right) separation of the two wrists is the biacromial *shoulder width*:

$$w_{\text{sh}} = \big\langle\, |b_\ell[0] - b_r[0]| \,\big\rangle_{\text{window}}. \tag{2}$$

3. Open hand + pinch. With the hands raised in front and opening/closing, each finger's bone length is the sum of its rigid phalanx segments—*curl-invariant*, so averaging across the open/closed sweep is valid and simultaneously yields the pinch open/close span.

2.3 Derived anthropometry

Let b_s^T be the T-pose wrist body coordinate for side $s \in \{\ell, r\}$. The arm length is the lateral distance from the shoulder ($\pm w_{\text{sh}}/2$) to the wrist,

$$L^s_{\text{arm}} = \big|b_s^T[0]\big| - \tfrac{1}{2} w_{\text{sh}}, \qquad L_{\text{arm}} = \tfrac{1}{2}\big(L^\ell_{\text{arm}} + L^r_{\text{arm}}\big). \tag{3}$$

The shoulder-joint centre is stored as a fixed body-frame offset (rebuilt live from the current head pose for the shoulder anchor),

$$c^s_{\text{sh}} = \Big(\pm \tfrac{1}{2} w_{\text{sh}},\ \big\langle b_s^T[1], b_s^F[1] \big\rangle,\ b_s^T[2] \Big), \qquad h_{\text{neck}} = -\big\langle c^s_{\text{sh}}[1] \big\rangle_s, \tag{4}$$

where b_s^F is the arms-forward wrist coordinate (its height averaged with the T-pose for robustness) and h_{neck} is the eye-line height above the shoulder. The (frame-free) stature is recovered from the measured arm length via the Winter ratio $L_{\text{arm}} = 0.332\,H$:

$$H = L_{\text{arm}}/0.332. \tag{5}$$

Missing measurements fall back to ratios ($w_{\text{sh}} = 0.259\,H$, $h_{\text{neck}} = 0.118\,H$); per the NASA-STD-3000 percentile caveat, every directly measurable quantity is measured and ratios are used only for untracked segments.

2.4 From profile to control

Workspace scale (clutch mode, default). The clutch mapping's amplification (`RETARGET_SCALE` > 1) is deliberate—the operator moves the hands in a small comfortable box and the lazy base/torso/lift followers carry the large reaches—so the calibration *corrects* the tuned gain rather than replacing it. The correction is normalised to a reference operator arm length L_{ref}, so a reference-sized operator keeps the exact tuned feel:

$$s_{\text{clutch}} = \texttt{RETARGET_SCALE} \cdot \text{clip}\Big(\frac{L_{\text{ref}}}{L_{\text{arm}}},\, \gamma_{\min}, \gamma_{\max}\Big). \tag{6}$$

Shoulder-anchored mode (optional, absolute). Behind the `RETARGET_ANCHOR` flag, the palm target is anchored at the calibrated shoulder with an absolute 1:1 reach match (ACE-style [3]),

$$p_{\text{ee}}^{\text{robot}} = c_{\text{sh}}^{\text{robot}} + \gamma\, M_{\text{avp}\to\text{robot}}\big(p_{\text{wrist}} - c_{\text{sh}}^{\text{human}}\big), \qquad \gamma = \text{clip}\Big(\frac{r_{\text{robot}}}{L_{\text{arm}}}, \gamma_{\min}, \gamma_{\max}\Big), \tag{7}$$

with r_{robot} the rest-pose robot arm reach (shoulder body $\to$ palm body, computed live by FK) and $c_{\text{sh}}^{\text{human}}$ rebuilt from Eq. (4) at the live head pose. Orientation is passed $1:1$ in both modes (a rotation has no length to scale).

Per-finger sizing. The keypoint-IK retargeter scales each human finger about its knuckle to the robot bone. With calibration it uses the operator's *measured* canonical bone length ℓ_f^{cal} in place of a live per-frame estimate, freezing the per-finger ratio $r_f = \ell_f^{\text{robot}}/\ell_f^{\text{cal}}$. The grasp curve uses the operator's measured pinch span $(d_{\text{open}}, d_{\text{close}})$. An alternative AnyTeleop-style *vector* retargeter [5] matching relative keyvectors $\min_q \sum_i \|\alpha v_i^{\text{h}} - f_i(q)\|^2 + \beta\|q - q_-\|^2$ is provided (`hand_retargeting/vector.py`) for the comparison in §3; it is scale-invariant by construction.

3 Experiments

All figures and numbers are regenerated from scratch by `scripts/tools/calibration_experiments.py` (seeded; synthetic operators from the forward model in `calibration.py` run back through the real measurement pipeline).

3.1 Dimension recovery vs. tracker noise

Fig. 1 sweeps isotropic tracker noise and reports the mean absolute error of the recovered arm length, shoulder width, and neck height over $N = 25$ operators per level. Recovery is exact without noise and degrades gracefully: at a realistic 3 mm of per-joint noise the errors are 0.4 mm (arm), 1.0 mm (shoulder), and 0.7 mm (neck); at 5 mm they remain ≤ 1.8 mm (Table 1). Arm length is the most robust because it is a difference of window-averaged positions.

Table 1: Recovered-dimension error (mm, mean over operators) vs. per-joint tracker noise.

noise (mm, std)	0	1	2	3	5
arm length	0.0	0.1	0.2	0.4	0.6
shoulder width	0.0	0.3	0.7	1.0	1.8
neck height	0.0	0.2	0.4	0.7	1.2

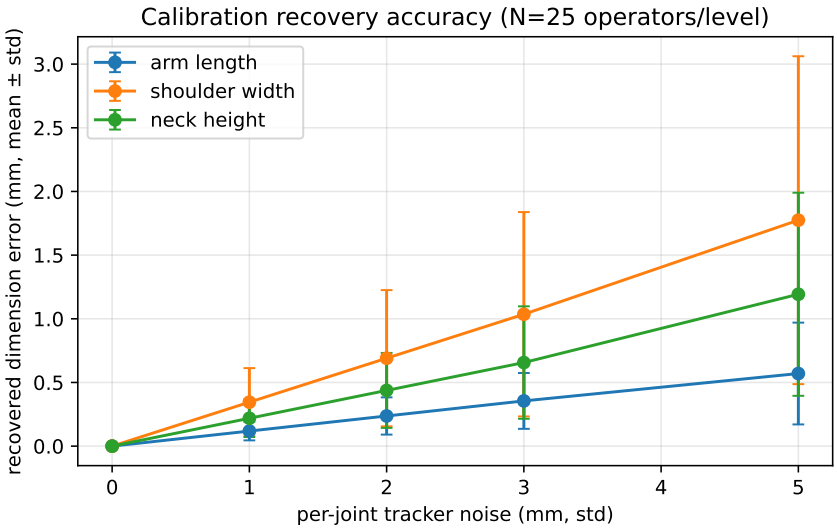


Figure 1: Calibration recovery accuracy vs. tracker noise.

3.2 Workspace-coverage consistency

Fig. 2 is the headline benefit. The effective workspace coverage (how far the robot palm target moves for a full operator arm-reach hand motion) is plotted across operators of 1.45 m–2.07 m. The *fixed* gain over/under-drives by 35.2 % across this range, so the robot feels different for each operator; the per-user clutch correction (6) and the absolute shoulder match (7) both hold coverage flat (0.0 % spread, the latter pinned exactly to the robot's 0.521 m reach).

3.3 Finger retargeter comparison

Fig. 3 compares the calibrated-length keypoint-IK against the vector optimizer across hand sizes (0.75×–1.3× nominal). Both solve in $\sim$ 1 ms (0.95 ms keypoint, 1.05 ms vector). By the absolute-fingertip metric—which the keypoint objective directly minimises—the calibrated keypoint solver is the more accurate (mean fingertip error 0.24 vs. 2.17 in normalised palm units); the vector method trades absolute-position accuracy for intrinsic scale-invariance, so the metric is biased toward the keypoint objective. This is why `config.FINGER_RETARGET` defaults to `keypoint` with calibrated lengths; the vector path is retained for cross-embodiment robustness studies.

4 Implementation & Usage

The calibrator consumes the frontend-agnostic `OperatorPose` (the same stream the teleop apps send), so it needs no new tracking. A typical session:

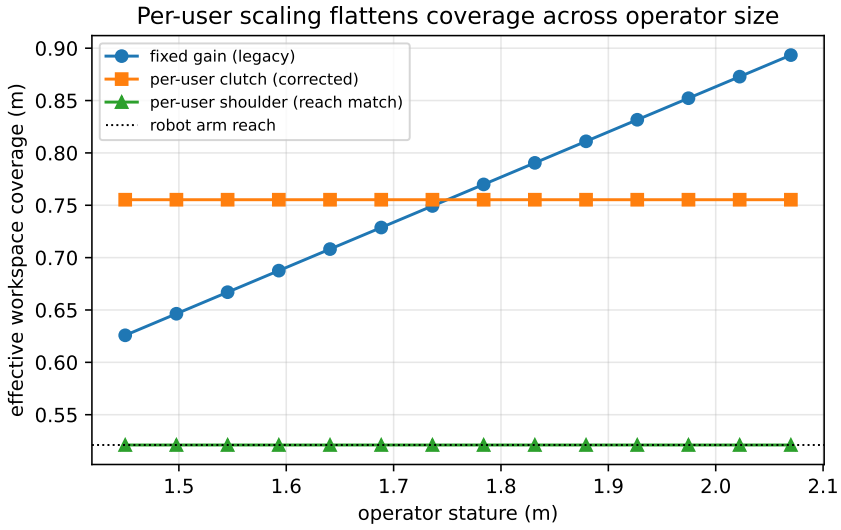


Figure 2: Per-user scaling flattens workspace coverage across operator size; the fixed gain varies by 35 %.

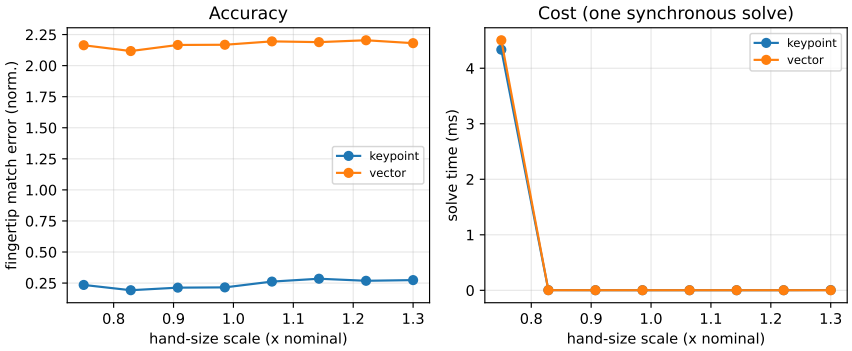


Figure 3: Finger retargeter A/B: accuracy (left) and per-solve cost (right) vs. hand size.

```
teleop = MabelTeleop(model, data)
teleop.load_profile("operator")          # apply a cached profile if present, else:
teleop.start_body_calibration("operator")# guided 3-pose capture; robot held still
while teleop.body_calibrating:           # each update() feeds the calibrator
    teleop.update(pose, dt=dt)           # profile applied + saved on completion
while running:                           # normal teleop, now per-user calibrated
    teleop.update(pose, dt=dt)
```

The profile parameterises `retgt.set_profile` (scale, shoulder anchor, pinch), `hand_retgt.set_profile` (per-finger lengths), and is cached to `profiles/<user>.json`. Set `config.ANTHRO_ARM_REF` to your own measured arm length so "unchanged feel" is anchored to you. The streaming math is exercised offline by `tests/test_calibration.py` (8 tests; noise-free round-trip, yaw invariance, finger recovery, γ clamping, shoulder offset, noisy robustness, the state machine, and JSON round-trip).

5 Limitations

- **Sparse 3-point under-constraint.** The shoulder-joint centre is inferred from anthropometric geometry, not measured directly; its error feeds the shoulder-anchored mode (7), which is therefore offered behind a flag with a head-anchored/clutch fallback. The default clutch mode (6) depends only on the robustly-measured arm length.
- **Stature from the headset.** The HMD eye-height stature estimate needs a floor-referenced world origin and is kept only as a secondary value; the primary stature is derived from the (frame-free) arm length.
- **Synthetic validation.** The experiments use a forward model with isotropic Gaussian noise; real ARKit hand tracking has structured, pose-dependent error and the operator's poses are imperfect (non-horizontal T-pose, non-parallel arms). On-hardware validation against a tape-measure ground truth is future work.
- **A-pose vs. T-pose, 1:1 vs. scaled.** The teleop literature lacks rigorous ablations of calibration pose and scale level specifically for VR humanoid arm teleop; the adjustable `ANTHRO_ARM_REF` and `RETARGET_ANCHOR` are the knobs for such a study.

References

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[6] D. A. Winter, *Biomechanics and Motor Control of Human Movement*, 4th ed., Wiley, 2009 (segment anthropometry, Fig. 4.1).